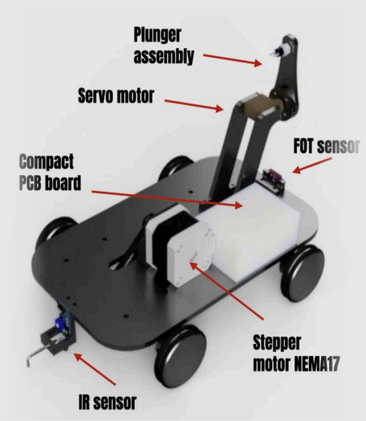
ENGINEERING & ENTERPRISE

My Project Portfolio

Selected university projects, from autonomous robotics to a student mobility start-up

3 projects
IMechE Design Challenge 2025 & 2026
Innovation & Enterprise (MED_6_IAE)

1



IMECHE DESIGN CHALLENGE 2025

Automated EV Charging Device

A fully autonomous, self-correcting device run by a Raspberry Pi. It records its forward travel after striking the first wall, then uses sensors to find the horizontal targets, while a servo-driven plunger reaches the vertical target. A NEMA17 stepper drives 3D-printed wheels fitted with rubber bands to prevent slipping.

0.1185 m/s
Velocity

2.63 mNm
Torque

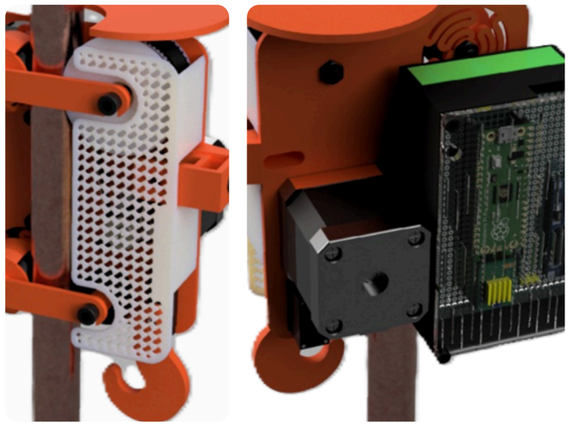
279.54 mm
Length (184 mm wide)

SKILLS & TECHNOLOGIES

Raspberry Pi · NEMA17 stepper · Servo motor · IR sensing · Pulley-belt drive · 3D printing · Circuit design

Team Vroom Vroom · My role: [add your contribution]

2



IMECHE DESIGN CHALLENGE 2026

Autonomous Pipe Climber

A four-wheeled autonomous climber that travels vertically on a 22 mm copper pipe. A stepper motor gives precise, repeatable movement, and a 3D-printed flexure spring clamping system keeps constant contact with the pipe to improve grip, stability and alignment. The result is a lightweight, compact and fully self-contained mechanism.

22 mm
Copper pipe

4-wheel
Clamped drive

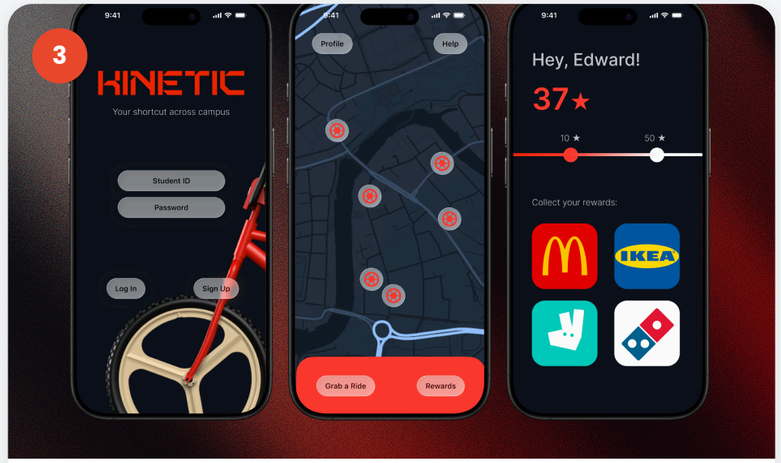
Stepper
Motor driven

SKILLS & TECHNOLOGIES

Flexure spring design · Stepper motor · 3D printing · Mechanism design · Autonomous systems

Team Vroom Vroom · My role: [add your contribution]

3



INNOVATION & ENTERPRISE · MED_6_IAE

Kinetic: Student E-Bike Sharing

A business concept for a fully electric, tap-and-go bike scheme for university students, with pay-per-km pricing, student discounts and a rewards app. The team pitch covers competitor analysis, market sizing, financial projections, a roadmap and risk mitigation.

£1.24M
Year 1 revenue (proj.)

£821k
Year 1 profit (proj.)

£730.8M
Target available market

SKILLS & TECHNOLOGIES

Business modelling · Market sizing · Competitor analysis · App concept design · Risk assessment · Pitching

My role: [add your contribution]

NAME
[Your Name]

EMAIL
[your.email@example.com]

LINKEDIN
[linkedin.com/in/yourname]

GITHUB
[github.com/yourname]